

# Dissertation

## A Distributed Edge FLISR Solution & Network Simulation Test Platform

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## 1 Introduction

Similar style and content as the proposal document, possibly expand it out where appropriate.

### 1.1 Research background

### 1.2 Motivations for the research

### 1.3 Research objectives

## 2 Review of the literature

Similar style and content as the proposal document, expand out into more research topics (bring in simulations?)

## **2.1 Digitalisation of the grid**

## **2.2 Fault Location, Isolation & Service Restoration**

## **2.3 Edge computing**

# **3 Methods**

Methods section will describe the context of the research and the actual work done throughout the dissertation.

## **3.1 Data & Setup of Experiment Environment**

This section will describe the underlying data and work of setting up the experiment environment from the simulation and FLISR Edge perspective.

### **3.1.1 Trial grid topology features**

Describe the trial sites selected to use for simulations and testing, describe the features (network structure, no. of switches etc.)

### **3.1.2 FLISR Edge solution & algorithm**

Describe the FLISR Edge solution, how it is structured, orchestrated, the edge devices, algorithms etc.

## **3.2 Experiments**

This section will describe in detail the actual experiments undertaken with the simulation platform and FLISR algorithms.

### **3.2.1 Network simulation test platform**

Describe simulation platform and its structure, purpose, how it operates etc.

### **3.2.2 Fault simulation operation**

Describe how the FLISR Edge solution and simulation platform interact and operate, how it generates results and how these are presented.

# **4 Results of the research**

Section describing the results gathered from the testing, including the findings, relevance of the results to the DSO and consumer, and future work to expand the solution further.

## **4.1 Research findings**

Describe the findings, how effective the solution is (or not), compare and contrast across the selected trial grids, centralised vs. distributed, limitations etc.,

## **4.2 Relevance to the DSO**

Describe the relevance of the results to the DSO and the consumer, from a grid stability, financial and climate perspective (weather events).

## **4.3 Future work**

## **5 Conclusion**

## **6 Exploitation Activities**

Describe any exploitation activities (papers/presentations etc.)

## References